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JEE Main Physics Practice Sheet — DPP-036

Topic: Work Done by a Variable Force — Graphical & Integration Methods
Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

- Q1.** A position-dependent force $F(x) = (3x^2 - 2x + 5)$ N acts on a particle moving along the x -axis. The work done by the force in displacing the particle from $x = 0$ to $x = 2$ m is:
- (A) 14 J
(B) 18 J
(C) 12 J
(D) 20 J
- Q2.** The force-displacement (F - x) curve for a particle consists of a straight line increasing from $F = 0$ at $x = 0$ to $F = 20$ N at $x = 4$ m, and remaining constant at 20 N from $x = 4$ m to $x = 8$ m. The total work done from $x = 0$ to $x = 8$ m is:
- (A) 80 J
(B) 120 J
(C) 160 J
(D) 100 J
- Q3.** A spring of spring constant $k = 400$ N/m is initially stretched by $x_1 = 5$ cm. The additional work required to stretch it further to $x_2 = 15$ cm is:
- (A) 2.0 J
(B) 8.0 J
(C) 4.0 J
(D) 1.0 J
- Q4.** A variable force $\vec{F} = (2x\hat{i} + 3y^2\hat{j})$ N acts on a body displacing it from position (1 m, 2 m) to (3 m, 4 m). The work done by this force is:
- (A) 56 J
(B) 60 J
(C) 64 J
(D) 48 J
- Q5.** A force acting on a particle varies with position x as $F(x) = F_0 \sin\left(\frac{\pi x}{L}\right)$. The work done by this force in moving the particle from $x = 0$ to $x = L$ is:
- (A) $\frac{2F_0 L}{\pi}$
(B) Zero
(C) $\frac{F_0 L}{\pi}$
(D) $\frac{F_0 L}{2\pi}$
- Q6.** A particle of mass $m = 2$ kg is accelerated from rest along the x -axis such that its velocity varies with displacement as $v = 3\sqrt{x}$ m/s. The work done by the net force acting on the particle during its displacement from $x = 0$ to $x = 4$ m is:
- (A) 72 J
(B) 36 J
(C) 18 J
(D) 144 J
- Q7.** In an F - x graph, the area of the region above the displacement axis represents +30 J of work, while a triangular region below the displacement axis has a base of 4 m and height -10 N. The net work done during the complete process is:
- (A) 50 J
(B) 10 J
(C) 20 J
(D) -10 J
- Q8.** A non-linear elastic spring exerts a restoring force $F(x) = -(ax + bx^3)$, where $a = 100$ N/m and $b = 400$ N/m³. The work done by an external agent to compress the spring by $x = 0.1$ m from its natural length is:
- (A) 0.50 J
(B) 0.60 J
(C) 0.51 J
(D) 0.54 J
- Q9.** A particle moves in the xy -plane under the force $\vec{F} = (y\hat{i} + x\hat{j})$ N. The work done in moving the particle from (0, 0) to (2 m, 3 m) along any arbitrary path is:
- (A) 5 J
(B) 12 J
(C) 6 J
(D) Zero
- Q10.** A body of mass $m = 0.5$ kg moves along the x -axis subject to a force $F(x) = -kx$, where $k = 50$ N/m. If the body is released from $x = 0.2$ m with zero velocity, its kinetic energy when it reaches $x = 0.1$ m is:
- (A) 0.25 J
(B) 1.00 J
(C) 0.75 J
(D) 0.50 J
- Q11.** A force $F = \frac{c}{x^2}$ (where $c = 12$ N·m²) acts on a particle on the x -axis. The work done by this force in moving the particle from $x = 2$ m to $x = 6$ m is:
- (A) 4 J
(B) 6 J
(C) 2 J

(D) 8 J

Q12. A chain of mass M and length L lies on a smooth horizontal tabletop with a fraction η ($0 < \eta < 1$) of its length hanging freely over the edge. The work required to pull the hanging part completely back onto the table is:

(A) $\frac{1}{2}\eta^2 MgL$

(B) ηMgL

(C) $\frac{1}{2}\eta MgL$

(D) $\eta^2 MgL$

Q13. A force $\vec{F} = -k(y\hat{i} - x\hat{j})$ N acts on a particle. The work done by this force as the particle completes one full revolution along a circle of radius R in the counterclockwise direction in the xy -plane is:

(A) Zero

(B) $2\pi kR^2$

(C) πkR^2

(D) $-2\pi kR^2$

Q14. A particle of mass $m = 1$ kg moves along the x -axis under a conservative force whose potential energy is given by $U(x) = (4x^2 - 16x + 20)$ J. The work done by the force as the particle moves from $x = 1$ m to $x = 3$ m is:

(A) Zero

(B) +8 J

(C) -8 J

(D) +16 J

Q15. The displacement x of a particle of mass $m = 1$ kg as a function of time is given by $x = \frac{1}{3}t^3$, where x is in meters and t is in seconds. The work done by the net force acting on the particle in the first 2 seconds is:

(A) 4 J

(B) 8 J

(C) 16 J

(D) 32 J

Q16. A liquid of density ρ fills a cylindrical container of cross-sectional area A to height H . The minimum work done to pump all the liquid out through a small hole at the top of the container is:

(A) $\frac{1}{2}\rho gAH^2$

(B) ρgAH^2

(C) $\frac{1}{3}\rho gAH^2$

(D) $\frac{1}{4}\rho gAH^2$

Q17. A force $F(x) = \alpha x e^{-\beta x^2}$ acts on a particle along the x -axis. The total work done by this force as the particle moves from $x = 0$ to $x \rightarrow \infty$ is:

(A) $\frac{\alpha}{\beta}$

(B) $\frac{\alpha}{2\beta}$

(C) $\frac{2\alpha}{\beta}$

(D) $\frac{\alpha}{4\beta}$

Q18. A block is connected to two identical springs in parallel, each of stiffness k . If an external agent stretches the combination by an extension x from its natural length, the work done on the spring system is:

(A) $\frac{1}{2}kx^2$

(B) kx^2

(C) $2kx^2$

(D) $\frac{1}{4}kx^2$

Q19. A particle moves along the curve $y = 2x^2$ from $(0, 0)$ to $(1 \text{ m}, 2 \text{ m})$ under the action of force $\vec{F} = (3x^2\hat{i} + 2y\hat{j})$ N. The work done by the force is:

(A) 7 J

(B) 3 J

(C) 5 J

(D) 9 J

Q20. An elastic rubber band of natural length L requires a force $F(x) = ax + bx^2$ for an extension x . The work required to stretch the band by length L from its relaxed state is:

(A) $\frac{1}{2}aL^2 + \frac{1}{3}bL^3$

(B) $aL^2 + bL^3$

(C) $\frac{1}{2}aL^2 + \frac{1}{2}bL^3$

(D) $\frac{1}{3}aL^2 + \frac{1}{4}bL^3$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-036)

Work Done by a Variable Force — Graphical & Integration Methods

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	A	6	B	11	A	16	A
2	B	7	B	12	A	17	B
3	C	8	C	13	B	18	B
4	C	9	C	14	A	19	C
5	A	10	C	15	B	20	A

Detailed Step-by-Step Solutions

Sol 1. Option (A)

Using 1D integration:

$$W = \int_0^2 (3x^2 - 2x + 5) dx = [x^3 - x^2 + 5x]_0^2 = (8 - 4 + 10) - 0 = 14 \text{ J}$$

Sol 2. Option (B)

Work done is the area under the F - x graph:

W = Area of triangle ($0 \rightarrow 4$ m) + Area of rectangle ($4 \rightarrow 8$ m)

$$W = \left(\frac{1}{2} \times 4 \times 20\right) + (8 - 4) \times 20 = 40 + 80 = 120 \text{ J}$$

Sol 3. Option (C)

Work done against spring restoring force:

$$W = \frac{1}{2}k(x_2^2 - x_1^2) = \frac{1}{2}(400) [(0.15)^2 - (0.05)^2]$$

$$W = 200 \times [0.0225 - 0.0025] = 200 \times 0.02 = 4.0 \text{ J}$$

Sol 4. Option (C)

Integrating components independently:

$$W = \int_1^3 2x dx + \int_2^4 3y^2 dy = [x^2]_1^3 + [y^3]_2^4$$

$$W = (9 - 1) + (64 - 8) = 8 + 56 = 64 \text{ J}$$

Sol 5. Option (A)

Integrating the sinusoidal force:

$$W = \int_0^L F_0 \sin\left(\frac{\pi x}{L}\right) dx = F_0 \left[-\frac{L}{\pi} \cos\left(\frac{\pi x}{L}\right)\right]_0^L$$

$$W = -\frac{F_0 L}{\pi} [\cos \pi - \cos 0] = -\frac{F_0 L}{\pi} (-1 - 1) = \frac{2F_0 L}{\pi}$$

Sol 6. Option (B)

At $x = 0$, $v_i = 0$. At $x = 4$ m, $v_f = 3\sqrt{4} = 6$ m/s.

By the Work-Energy Theorem:

$$W = \Delta K = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = \frac{1}{2}(2)(6^2) - 0 = 36 \text{ J}$$

Sol 7. Option (B)

Net work is the algebraic sum of the signed areas:

$$W_{\text{net}} = \text{Area}_{\text{above}} - \text{Area}_{\text{below}}$$

$$\text{Area}_{\text{below}} = \frac{1}{2} \times \text{base} \times |\text{height}| = \frac{1}{2} \times 4 \times 10 = 20 \text{ J}$$

$$W_{\text{net}} = 30 - 20 = 10 \text{ J}$$

Sol 8. Option (C)

Work done by external agent against the restoring force $F_{\text{ext}} = ax + bx^3$:

$$W = \int_0^{0.1} (100x + 400x^3) dx = [50x^2 + 100x^4]_0^{0.1}$$

$$W = 50(0.01) + 100(0.0001) = 0.50 + 0.01 = 0.51 \text{ J}$$

Sol 9. Option (C)

Notice that $dW = y dx + x dy = d(xy)$.

Because the differential is exact, the force is conservative:

$$W = \int_{(0,0)}^{(2,3)} d(xy) = [xy]_{(0,0)}^{(2,3)} = (2 \times 3) - 0 = 6 \text{ J}$$

Sol 10. Option (C)

By Work-Energy Theorem:

$$\Delta K = W_{\text{spring}} = -\int_{0.2}^{0.1} 50x dx = [-25x^2]_{0.2}^{0.1}$$

$$K_f - 0 = -25(0.01 - 0.04) = -25(-0.03) = 0.75 \text{ J}$$

Sol 11. Option (A)

Integrating the inverse-square force:

$$W = \int_2^6 \frac{12}{x^2} dx = \left[-\frac{12}{x}\right]_2^6 = -\frac{12}{6} - \left(-\frac{12}{2}\right) = -2 + 6 = 4 \text{ J}$$

Sol 12. Option (A)

Mass of hanging part: $M_h = \eta M$.

Center of mass of the hanging portion is at depth: $y_{\text{cm}} = \frac{\eta L}{2}$.

Work required to lift center of mass onto table:

$$W = M_h g y_{\text{cm}} = (\eta M) g \left(\frac{\eta L}{2}\right) = \frac{1}{2} \eta^2 M g L$$

Sol 13. Option (B)

In polar coordinates: $x = R \cos \theta$, $y = R \sin \theta$, $dx = -R \sin \theta d\theta$, $dy = R \cos \theta d\theta$.

$$dW = -k(y dx - x dy) = -k[R \sin \theta (-R \sin \theta d\theta) - R \cos \theta (R \cos \theta d\theta)]$$

$$dW = kR^2(\sin^2 \theta + \cos^2 \theta) d\theta = kR^2 d\theta$$

Integrating over one full loop ($\theta = 0 \rightarrow 2\pi$):

$$W = \int_0^{2\pi} kR^2 d\theta = 2\pi kR^2$$

Sol 14. Option (A)

For a conservative force: $W = -\Delta U = -(U_f - U_i) = U_i - U_f$.

At $x = 1$ m: $U(1) = 4(1) - 16(1) + 20 = 8$ J.

At $x = 3$ m: $U(3) = 4(9) - 16(3) + 20 = 36 - 48 + 20 = 8$ J.

Work done: $W = U(1) - U(3) = 8 - 8 = 0$ J.

Sol 15. Option (B)

Velocity as a function of time: $v(t) = \frac{dx}{dt} = t^2$.

At $t = 0$, $v(0) = 0$. At $t = 2$ s, $v(2) = 2^2 = 4$ m/s.

By the Work-Energy Theorem:

$$W = \Delta K = \frac{1}{2}mv(2)^2 - 0 = \frac{1}{2}(1)(4^2) = 8 \text{ J}$$

Sol 16. Option (A)

Total mass of liquid: $M = \rho AH$.

Center of mass of the liquid column is at depth $h_{\text{cm}} = \frac{H}{2}$ below the top rim.

Work required to raise the entire mass out of the top:

$$W = Mgh_{\text{cm}} = (\rho AH)g\left(\frac{H}{2}\right) = \frac{1}{2}\rho gAH^2$$

Sol 17. Option (B)

Let substitution $u = \beta x^2 \implies du = 2\beta x dx \implies x dx = \frac{du}{2\beta}$.

$$W = \alpha \int_0^\infty x e^{-\beta x^2} dx = \alpha \int_0^\infty e^{-u} \frac{du}{2\beta} = \frac{\alpha}{2\beta} [-e^{-u}]_0^\infty = \frac{\alpha}{2\beta} (0 - (-1)) = \frac{\alpha}{2\beta}$$

Sol 18. Option (B)

Equivalent spring constant for two identical parallel springs:

$k_{\text{eq}} = k + k = 2k$.

Work done in stretching the parallel combination:

$$W = \frac{1}{2}k_{\text{eq}}x^2 = \frac{1}{2}(2k)x^2 = kx^2$$

Sol 19. Option (C)

Exact differential: $dW = 3x^2 dx + 2y dy = d(x^3 + y^2)$.

$$W = [x^3 + y^2]_{(0,0)}^{(1,2)} = (1^3 + 2^2) - 0 = 1 + 4 = 5 \text{ J}$$

Sol 20. Option (A)

Integrating from $x = 0$ to $x = L$:

$$W = \int_0^L (ax + bx^2) dx = \left[\frac{1}{2}ax^2 + \frac{1}{3}bx^3 \right]_0^L = \frac{1}{2}aL^2 + \frac{1}{3}bL^3$$